

AI-Driven Sustainable Energy Analytics and Renewable Energy Prediction Using Machine Learning

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Project Objectives

- Problem Statement
- Project Overview – Introduction
- End Users
- Wow Factor in Project
- Modelling/Block Diagram/Flow of Project
- Result/outcomes
- Conclusion
- Future Perspective



Problem Statement

Renewable energy sources such as solar and wind are highly dependent on environmental conditions. Accurate prediction of energy generation is essential for efficient power management.

Challenges:

- Unpredictable weather conditions.
- Large amounts of historical energy data.
- Difficulty in accurate forecasting.
- Inefficient utilization of renewable resources.
- High dependency on fossil fuels due to inaccurate predictions.

Proposed Solution

- EcoAI uses Artificial Intelligence and Machine Learning algorithms to analyze historical energy datasets and generate accurate renewable energy predictions.

Project Overview - Introduction

Introduction

- EcoAI is an AI-powered renewable energy analytics system designed to predict future renewable energy generation using Machine Learning.

Project Workflow

- Collect renewable energy dataset.
- Clean and preprocess the data.
- Perform Exploratory Data Analysis (EDA).
- Train Machine Learning models.
- Evaluate model performance.
- Generate renewable energy predictions.

Benefits

- Improves forecasting accuracy.
- Helps optimize energy production.
- Supports sustainable development.
- Enables data-driven decision-making.

End Users

Target Users

- EcoAI can be used by various organizations involved in renewable energy planning and management.

End Users

- Renewable Energy Companies
- Government Energy Departments
- Smart Grid Operators
- Environmental Researchers
- Educational Institutions
- Sustainability Organizations
- Data Scientists and Researchers
- Policy Makers

Benefits for Users

- Better energy planning.
- Cost reduction.
- Improved resource utilization.
- Environmental sustainability.

Wow Factors in Project

Unique Features

AI-Based Prediction

- Uses Machine Learning algorithms to predict renewable energy generation.

Data Visualization

- Provides meaningful charts and graphs for better understanding.

Automated Data Processing

- Performs preprocessing, feature selection, and prediction automatically.

Sustainable Development

- Supports clean energy initiatives and SDGs.

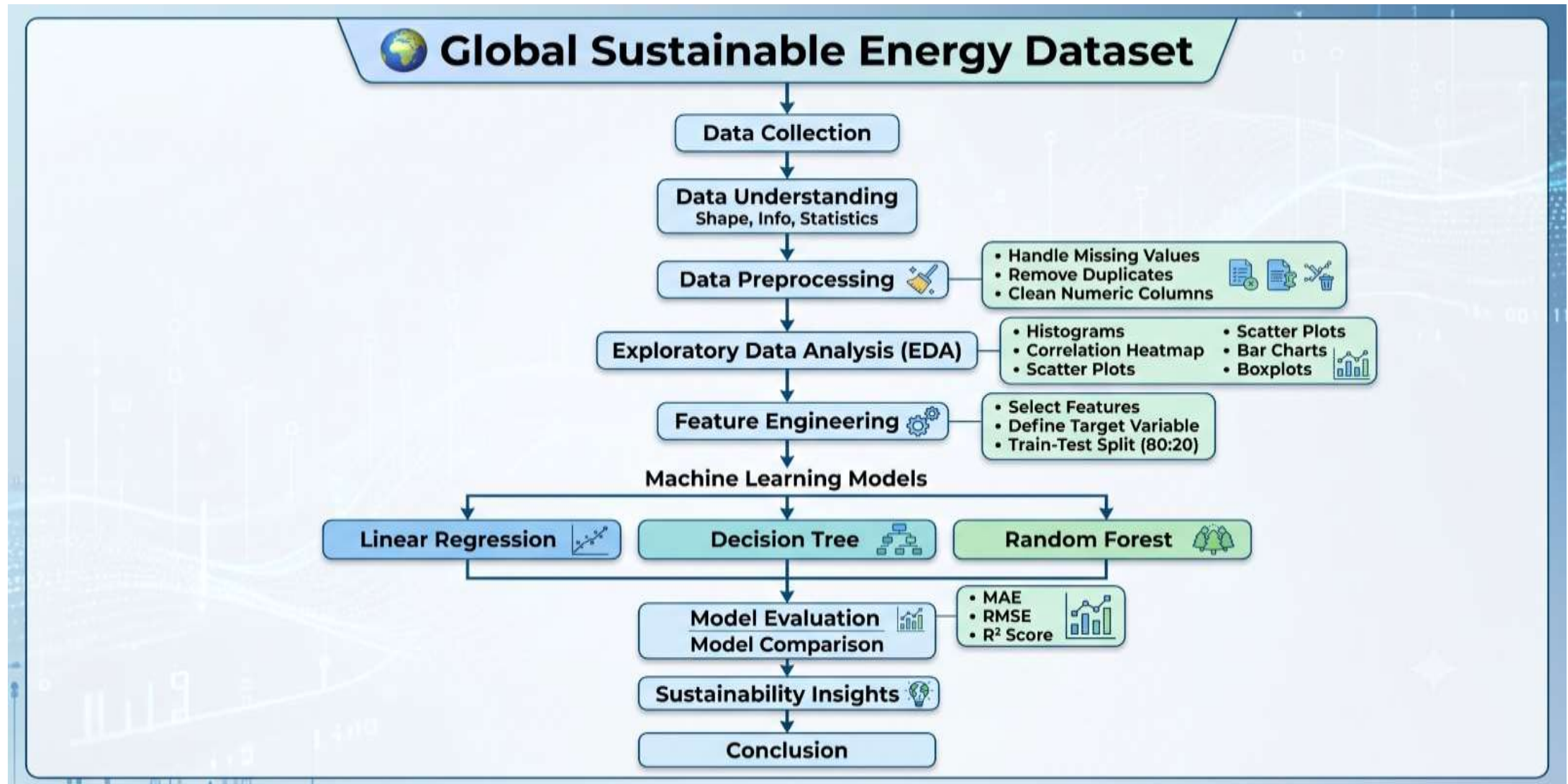
Scalable Solution

- Can be extended to real-time prediction systems.

Decision Support

- Helps governments and industries make informed energy decisions.

Modelling/Block Diagram/Flow of Project



Technology Stack

Software Used

Programming Language

- Python

Development Environment

- Jupyter Notebook

Libraries

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

Machine Learning

- Regression Algorithms
- Model Evaluation Metrics

Visualization

- Heatmaps
- Histograms
- Scatter Plots
- Correlation Matrix

Result/Outcomes

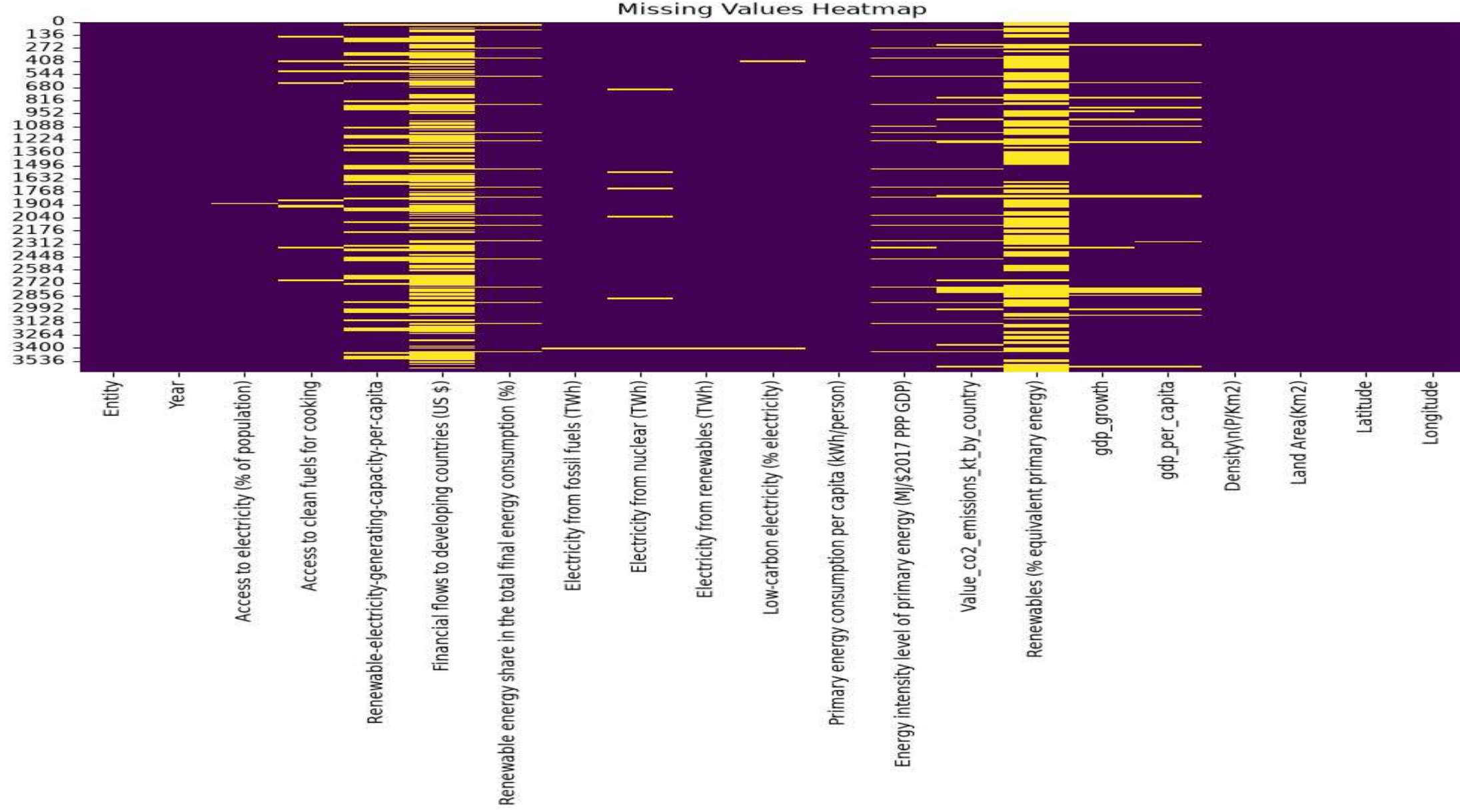
Project Results

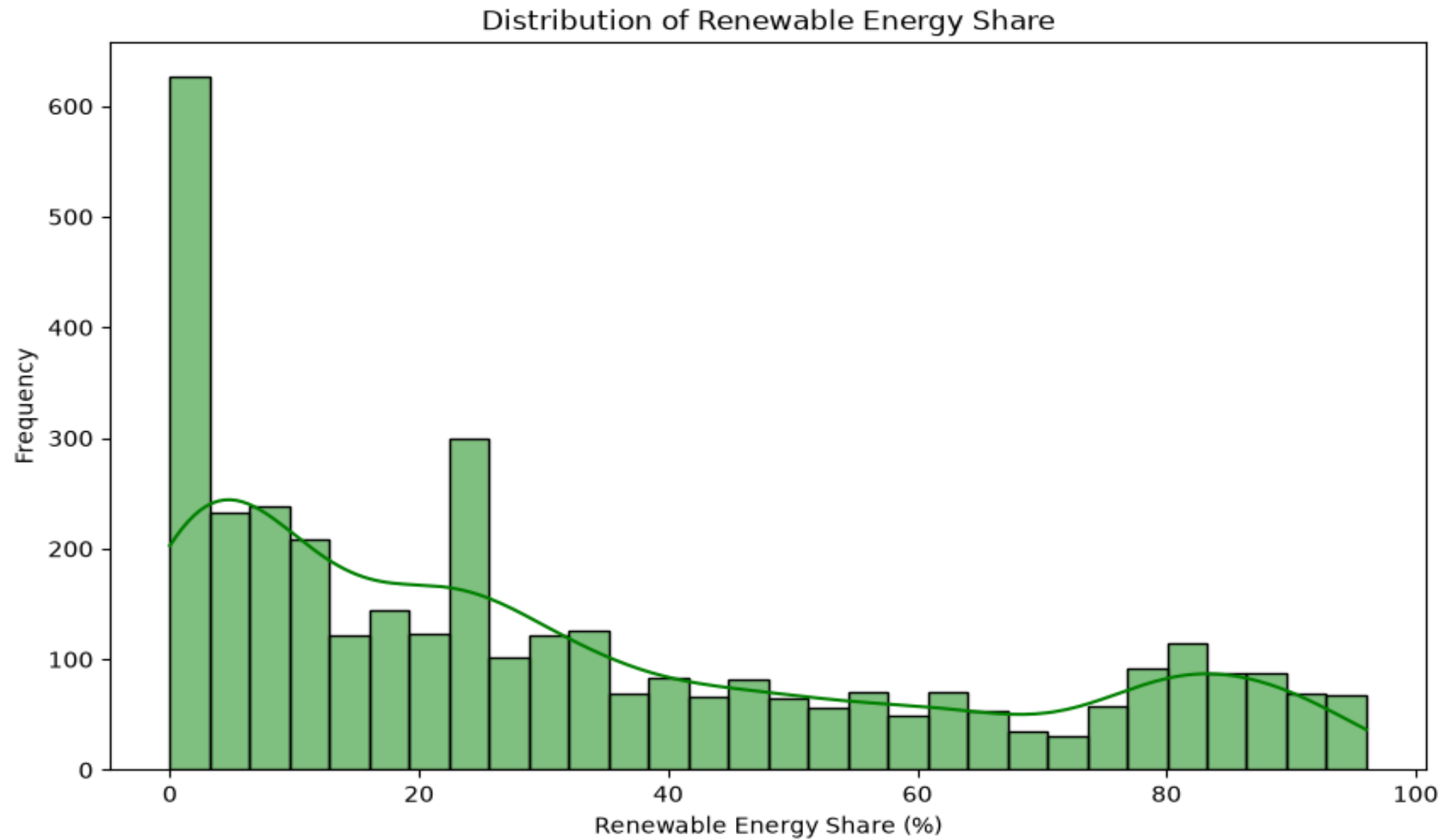
Achievements

- Successfully loaded and preprocessed the dataset.
- Performed Exploratory Data Analysis.
- Trained Machine Learning prediction model.
- Generated renewable energy predictions.
- Evaluated model performance.

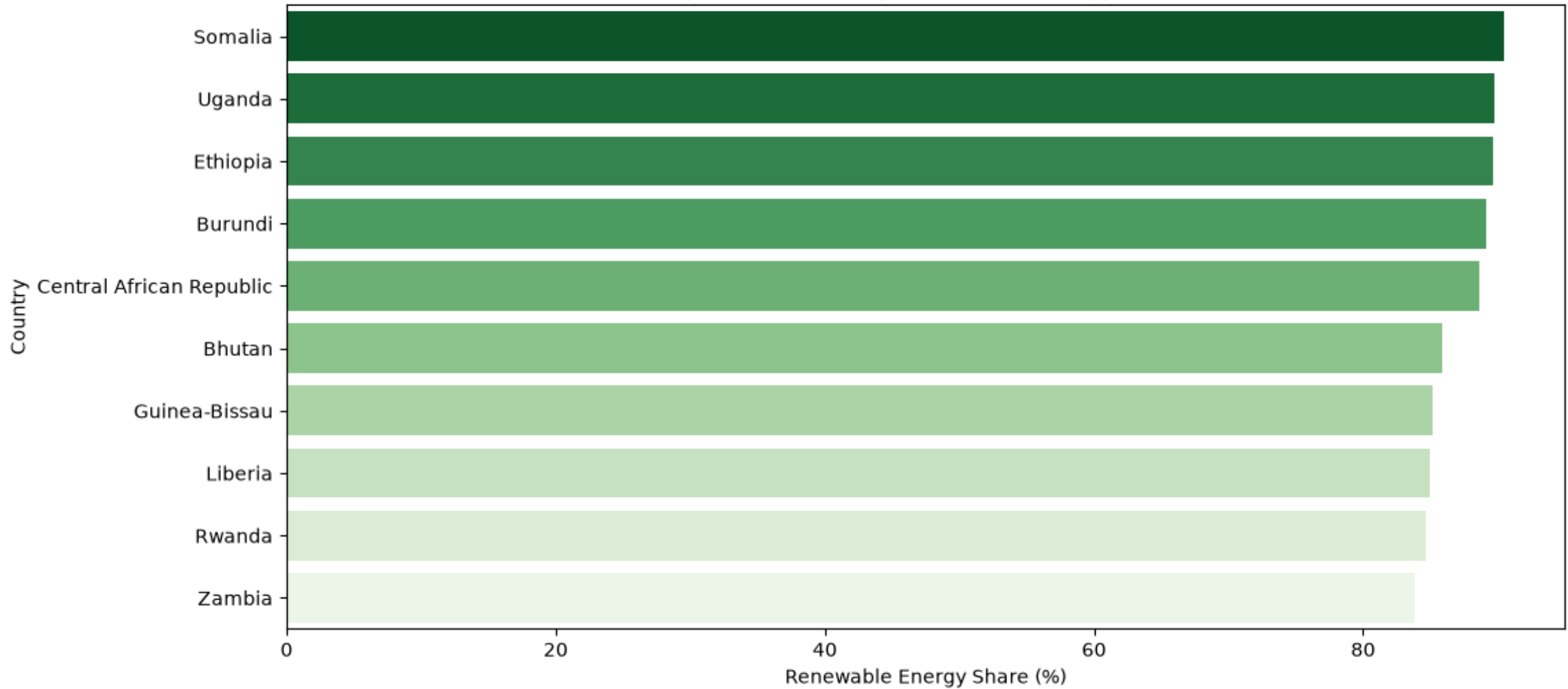
Outcomes

- Improved prediction accuracy.
- Better understanding of renewable energy trends.
- Effective data visualization.
- Reliable forecasting for sustainable planning.





Top 10 Countries by Renewable Energy Share



Conclusion

Conclusion:

- EcoAI demonstrates how Artificial Intelligence can significantly improve renewable energy forecasting through Machine Learning.

Key Highlights

- AI improves prediction accuracy.
- Supports efficient renewable energy utilization.
- Helps reduce dependence on fossil fuels.
- Encourages sustainable development.
- Provides valuable insights through data analysis.
- Supports informed decision-making for energy management.

Future Perspective

Future Scope

- The project can be further enhanced by integrating modern technologies

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Future Enhancements

- Real-time weather API integration.
- Deep Learning prediction models.
- IoT-based renewable energy monitoring.
- Cloud deployment.
- Mobile and Web Dashboard.
- Smart Grid integration.
- AI-powered energy optimization.
- Multi-source renewable energy forecasting.
- Real-time visualization dashboard.

Thank You

Github link:

https://github.com/umang-parmar/Edunate_Project

Video (google drive link)